

Construction and Comparison of Coronary Heart Disease Risk Prediction Models Based on Lasso-Logistic Regression and Random Forest Models

Zerui Jiang *

School of Public Health, Wuhan University, Wuhan, Hubei, 430072, China

* Corresponding Author Email: 2021303051008@whu.edu.cn

Abstract. Coronary heart disease (CHD) is a prominent global cause of death, including genetic and environmental factors. High CHD mortality rates are observed in South Africa and China, and its prevalence is increasing among younger demographics due to lifestyle changes. Topic This study investigates the relationship between CHD and its risk factors using cardiovascular data from Framingham, Massachusetts, to develop a predictive model. Methods The study processed data using R 4.4.1 software, dividing it into training (70%) and testing (30%) sets. Feature variables for the model were selected through univariate analysis and Lasso regression. Prediction models were constructed using multifactorial Logistic regression and random forest machine learning algorithms. Model performance was assessed using ROC curves and confusion matrices. Results Lasso regression and univariate Logistic regression analysis, as well as stepwise Logistic regression analysis, indicated that gender, age, daily smoking, stroke history, hypertension, cholesterol, systolic pressure, and glucose are significant CHD predictors. The multifactorial Logistic regression model achieved an 85.6% classification accuracy, 85.8% precision, 99.5% recall, 0.921 F1 Score, and 0.717 AUC. The random forest model showed slightly lower performance with 85.2% accuracy, 34.8% precision, 99.2% recall, 0.919 F1 Score, and 0.615 AUC. Conclusion This study not only screened the influencing factors of coronary heart disease but also found that the CHD risk prediction model constructed by machine learning algorithm has good predictive performance, especially the Logistic regression model performed better on the data set of this study.

Keywords: Coronary heart disease, Lasso regression, Logistic regression, Random Forest, Risk prediction model.

1. Introduction

Coronary heart disease (CHD) is one of the world's most common causes of death. Numerous risk factors, derived from both hereditary and environmental variables, contribute to the condition. According to the World Health Organization's (WHO) most recent data, in South Africa, about 56 people die from ischemic heart disease per 100,000 people, also known as coronary heart disease, ranking third among all causes of death. In China, this figure is as high as 123, ranking second [1].

Coronary heart disease seems to have become an epidemic in today's era, with high morbidity and mortality rates accompanied by a heavy economic burden. From 1990 to 2019, the global economic burden caused by coronary heart disease increased by 82%, reaching an average of 182 million dollars per year [2]. Overall, the incidence rate is increasing year by year, but the incidence rate in different countries can vary by as much as ten times [3]. Although many countries are gradually entering an aging population society, the mortality rate of congenital heart disease in some developed countries has dropped by 50% - this decline is said to be largely due to healthier lifestyles, better medical conditions, and unknown factors [4]. This indicates that there is a halt to the annual rise in the prevalence of coronary heart disease.

In clinical practice, many widely recognized indicators indicate potential risks in the diagnosis of coronary heart disease. Take hypertension for example, more than 200 million people are suffering from hypertension in China. Hypertension can not only induce coronary heart disease but also aggravate the condition of coronary heart disease, seriously affecting the quality of life of patients [5]. In the past, coronary heart disease mostly occurred in the elderly, but now, with lifestyle changes, more and more young people are beginning to show symptoms of coronary heart disease, which poses

a great threat to heart health. It seems necessary to consider age [6]. Smoking is a recognized independent risk factor for cardiovascular disease and death. It is strongly associated with unfavorable outcomes like high mortality and a higher risk of cardiovascular disorders like hypertension, coronary heart disease, and stroke. The greater the amount of smoking and the longer the smoking time, the higher the risk of cardiovascular disease and death [7]. Diabetes can damage blood vessels and increase the risk of coronary heart disease and stroke. In addition, diabetic patients often have high blood pressure and high cholesterol, which further increases the risk of disease [8]. Other factors such as gender, BMI, heart rate, blood sugar, and other easily obtainable indicators usually also play an important role in predicting CHD [9]. Many of these factors can be improved after information or intervention. The study INTERHEART showed that the risk of myocardial infarction depends largely on modifiable cardiovascular risk factors, and the same is true for coronary heart disease [10]. Therefore, early identification of reversible coronary heart disease risk factors and early intervention are extremely important.

This study analyzed an ongoing cardiovascular research dataset from residents of Framingham, Massachusetts, to explore the distribution characteristics of cardiovascular risk factors in patients with coronary heart disease and their correlation with coronary heart disease, aiming to further analyze their value in judging the clinical condition of patients.

2. Materials and Methods

2.1. Data Source

The analysis of all the data comes from cardiovascular research that was conducted on Framingham, Massachusetts, inhabitants [11]. The data mainly included general physical examination, laboratory tests, and medical history, and the final sample size included in the database was 4239 cases. This study is a component of the Framingham Heart Study (FHS), a ground-breaking research initiative that has spent the last 70 years identifying risk factors for cardiovascular disease and developing public health recommendations for its prevention. The FHS has consistently advanced at the forefront of epidemiological and cardiovascular science since its founding, guaranteeing the accuracy and dependability of the data.

In addition, the Framingham Heart Study data have been widely used in health economic modeling and have been validated for their accuracy in predicting cardiovascular outcomes. Although there have been concerns about its applicability in European populations, several studies have reported that the outcomes predicted by the Framingham risk score are generally accurate [12, 13]. This further strengthens the reliability of the data used in this study, as they come from an established and widely recognized source in the field of cardiovascular disease epidemiology.

2.2. Data Processing

The original data were screened, cleaned, and constructed into a new data set using R 4.4.1 software. We first eliminated data from the data set in the following steps: (1) Case deletion: eliminating cases with missing values exceeding 10%; (2) Data deletion: eliminating values that deviate significantly from the mean. The missing values were then filled in using the multiple regression interpolation method using the “mice” software package. The resulting new data set included 3,656 samples and 16 variables. The flow diagram is shown in Figure 1

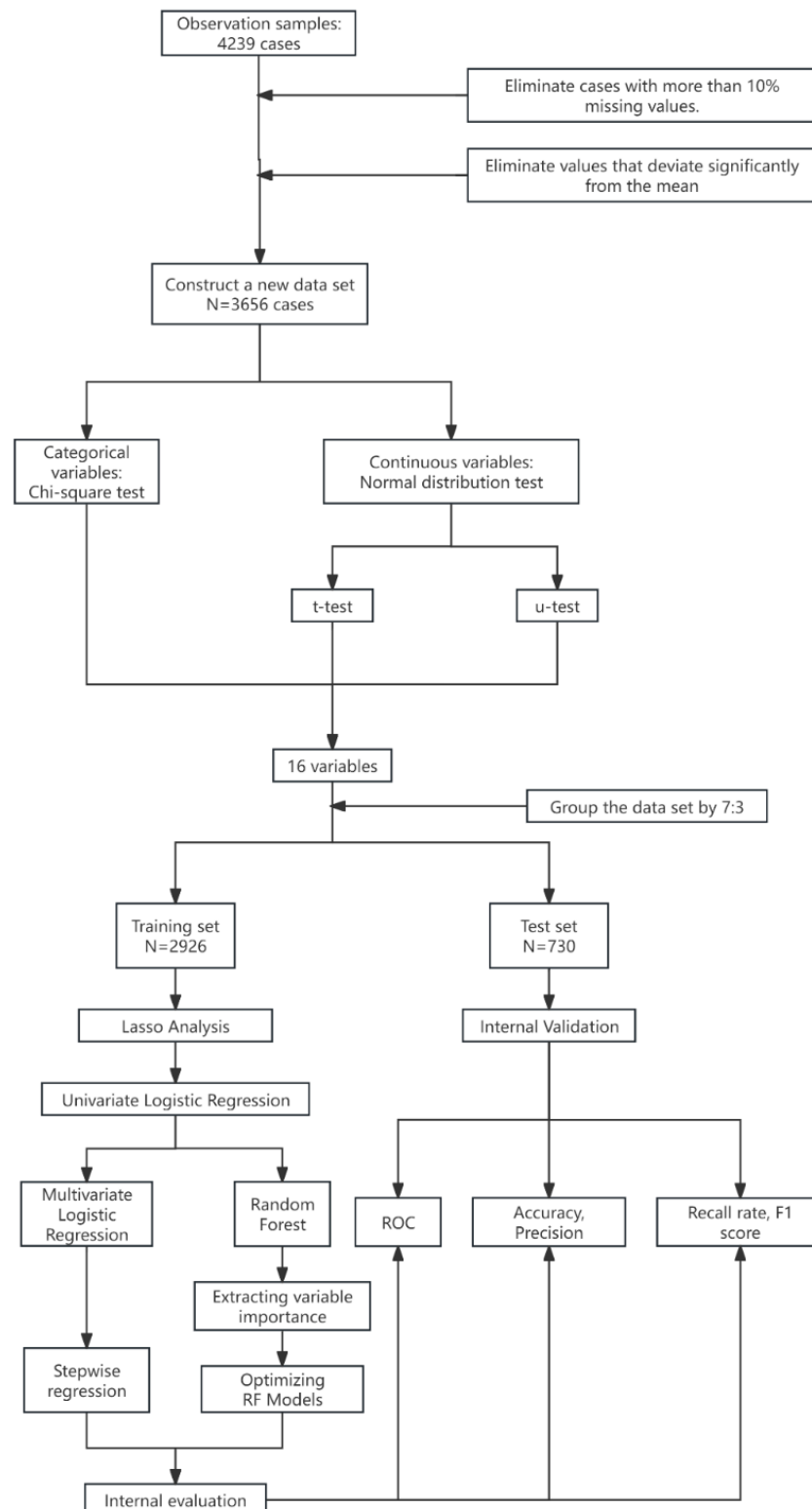


Figure 1. The flow diagram of data processing and model building (Picture credit: Original).

2.3. Univariate Analysis

The data was processed using R 4.4.1, and the categorical variables (gender, using blood pressure medication, having had a stroke, had hypertension, and had diabetes) were tested using chi-square tests. For continuous variables (such as age, smoking, total cholesterol, systolic and diastolic blood pressure, body mass index, heart rate, and glucose), T-tests and Mann-Whitney U tests were employed. The multivariate logistic regression analysis was used to examine the influential factors. P less than 0.05 was regarded as statistically significant.

2.4. Model Building

Following the completion of the preliminary study, the new data set was split at random (7:3) into a modeling group and a validation group. R software was then used to create a modeling group and a validation group for patients with coronary heart disease. The accuracy, precision, recall, F1 score, and AUC value of each model were determined by plotting its ROC curves and confusion matrices. These three models' data were finally compared.

3. Results

3.1. Univariate Analysis

15 independent variables, including gender, education level, and smoking status, were subjected to univariate analysis with the dependent variable being the patient's diagnosis of coronary heart disease. Gender, education level, BPMeds, Prevalent Stroke, Prevalent rHyp, Diabetes, age, cigsPerDay, totChol, sysBP, diaBP, BMI, and glucosae were found to differ statistically significantly ($P < 0.05$). The univariate analysis findings are displayed in Tables 1, 2, and 3.

Table 1. Results of Chi-square tests

Variable	P
Male	3.78e-08
Education	8.25e-07
CurrentSmoker	2.66e-01
BPMeds	1.49e-07
PrevalentStroke	8.82e-03
PrevalentHyp	8.46e-28
Diabetes	3.68e-08

Table 2. Results of u-tests

Variable	P	Test
Age	2.90e-44	Mann-Whitney
CigsPerDay	1.74-02	Mann-Whitney
TotChol	2.02-07	Mann-Whitney
SysBP	6.48e-33	Mann-Whitney
DiaBP	1.52-15	Mann-Whitney
BMI	1.58e-06	Mann-Whitney
HeartRate	4.04e-01	Mann-Whitney
Glucose	9.962163e-04	Mann-Whitney

Table 3. Results of t-tests

Variable	P	Test
Age	4.32e-45	T-test
CigsPerDay	3.26-03	T-test
TotChol	4.20e-07	T-test
SysBP	9.98e-28	T-test
DiaBP	2.50e-14	T-test
BMI	6.43e-06	T-test
HeartRate	2.20e-01	T-test
Glucose	5.06e-06	T-test

3.2. Data Splitting

In the R 4.4.1 software environment, the caTools package was used to divide the data set into two parts: a training set of 2926 cases (accounting for 70% of the total) and a validation set of 730 cases

(accounting for 30% of the total). The model was constructed using the training set, and it was assessed using the validation set.

3.3. Feature Variable Screening by Lasso Regression

The variables with a P value less than 0.05 in the univariate analysis, or the variables with significant differences, were included in the Lasso regression analysis with the dependent variable being whether to develop coronary heart disease 10 years later. Before the Lasso regression analysis, the features were standardized so that the mean of each feature was 0 and the standard deviation was 1, which improved the stability and convergence speed of the model. After training the Lasso model, the optimal solution of the parameter λ was obtained through cross-validation and the final model was trained using the optimal parameter λ . The visualized cross-validation results are shown in Figure 2. Finally, the coefficients were extracted and predicted. The coefficient extraction results are shown in Table 4.

Lasso regression findings are displayed in Table 4 and Figure 2, where Table 4 displays the coefficient values for each variable. The lambda. In in Figure 2 represents the lambda value that minimizes the mean square error (MSE) among all lambda values, which can ensure that the model has the best prediction performance. lambdas represent the lambda value within one standard error range of the model error. This study selects lambda. In as the optimal lambda value. The results show that when lambda (λ) is 0.001169531, the error is the smallest, and the number of influencing factors corresponding to it is 10. The corresponding coefficients are shown in Table 4. Among them, the coefficients of age and sysBP are large, indicating that they have a greater impact on the target variable, and the coefficients of education and diaBP are small (especially education is negative), indicating that they have a smaller impact on the target variable.

Table 4. Coefficient values corresponding to each variable

(Intercept)	0.152426521
male	0.027205435
age	0.061974525
education	/
cigsPerDay	0.029187592
BPMeds	0.001747651
prevalentStroke	0.010989029
prevalentHyp	0.010662105
diabetes	0.007209364
totChol	0.005341188
sysBP	0.038659696
diaBP	/
BMI	/
glucose	0.025388513

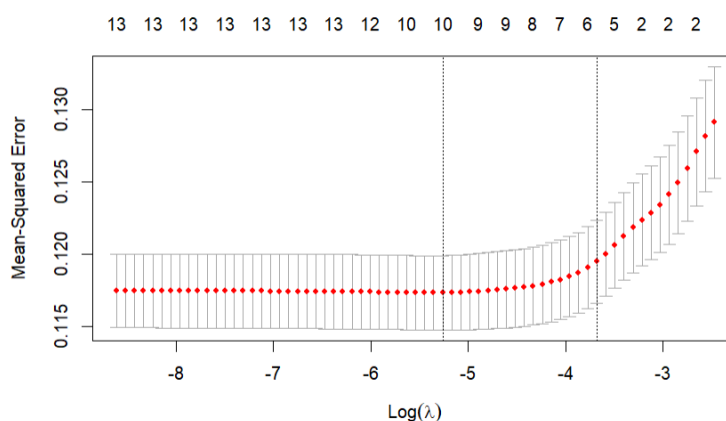


Figure 2. Plot of lambda value and mean square error (MSE) (Picture credit: Original).

3.4. Model Building

3.4.1. Multivariate Logistic Regression

The 10 characteristic variables screened by Lasso analysis were first subjected to univariate logistic regression. The results are shown in Table 5. A total of 10 characteristic variables were screened ($P < 0.05$). Then they were incorporated into the multivariate logistic analysis model. After the multivariate logistic regression model was constructed, the VIF value of each variable was calculated. The obtained VIF values are shown in Figure 3. The findings demonstrate that there is essentially no multicollinearity between the variables because all the VIF values fall between 1 and 5.

Table 5. Univariate logistic regression results

Variable	P
age	1.864311e-34
sysBP	3.329152e-28
prevalentHyp	1.696336e-20
glucose	1.151934e-09
male	2.230479e-08
diabetes	3.934300e-08
totChol	2.157724e-07
BPMeds	5.864524e-05
cigsPerDay	1.952284e-04
prevalentStroke	3.561289e-03

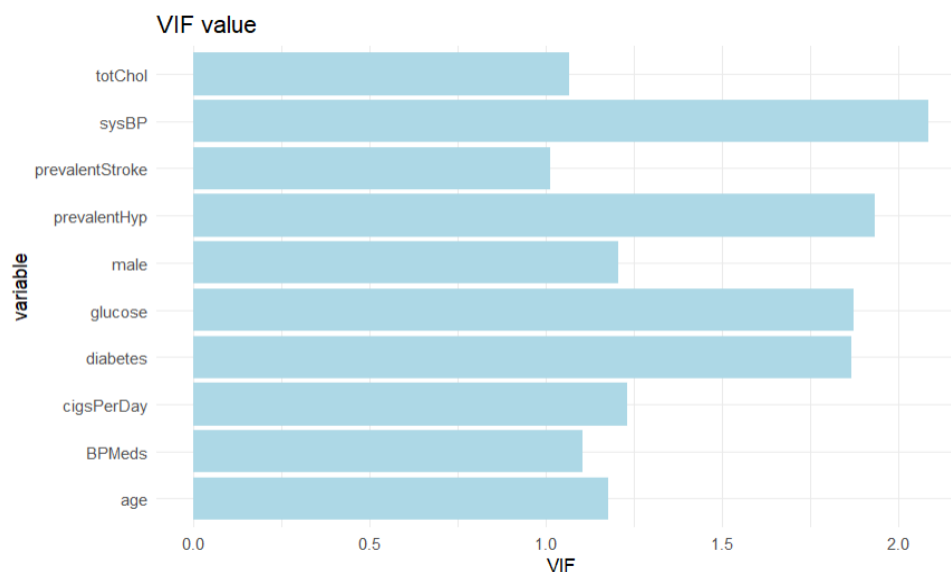


Figure 3. VIF value of each variable: used to detect multicollinearity (Picture credit: Original).

Table 6 displays the multivariate logistic regression analysis findings. There are several non-significant variables. To further simplify the model, this paper adopts the stepwise regression method to automatically select the most significant variables. The BPMeds variable was deleted in the first iteration, and the AIC of the model dropped from 2214.36 to 2212.42. Although other variables (such as diabetes) also reduce the AIC, the reduction is smaller. The second iteration removed the diabetes variable, and the AIC of the model further decreased to 2210.82. At this point, removing other variables did not further reduce the AIC. After excluding variables that have no significant contribution to model prediction, a total of 8 feature variables were screened out. They are male, age, cigsPerDay, preferable Stroke, preferableHyp, totChol, sysBP, and glucose. See Table 7. Although the P values of preferential Stroke and preferentially are greater than 0.05, they are still retained in the model because retaining them can significantly reduce the AIC value of the model.

The formula of the logistic regression model can be derived based on the coefficient estimates provided. To find the probability of coronary heart disease, substitute the predicted patient's data into the logarithmic probability. Next, use the sigmoid function, which is the logit function's inverse function, to convert the logit value back to the probability value.

$$\text{Logit}(p) = -8.916855 + 0.611539\text{Male} + 0.067382\text{Age} + 0.021991\text{CigsPerDay} + 0.988329\text{PrevalentStroke} + 0.239780\text{PrevalentHyp} + 0.002960\text{TotChol} + 0.012674\text{SySBP} + 0.007855\text{Glucose} \quad (1)$$

Table 6. Coefficients and P values of multivariate logistic regression analysis

Coefficients	Estimate	P
(Intercept)	-8.811407	<2e-16
male	0.612404	3.97e-07 ☆☆☆
age	0.067194	<2e-16 ☆☆☆
cigsPerDay	0.022033	1.67e-06 ☆☆☆
BPMeds1	0.064024	0.81199
prevalentStroke1	0.983867	0.07311
prevalentHyp1	0.234969	0.12470
diabetes1	0.219340	0.53009
totchol	0.002927	01777 ☆
sySBP	0.012600	9.95e-05 ☆☆☆
glucose	0.006810	0.00564 ☆☆

Table 7. The variables retained after stepwise regression and their corresponding coefficients

Coefficients	Estimate	P
(Intercept)	-8.916855	<2e-16
male	0.611539	3.98e-07 ☆☆☆
age	0.067382	<2e-16 ☆☆☆
cigsPerDay	0.021991	1.76e-06 ☆☆☆
prevalentStroke	0.988329	0.0707
prevalentHyp	0.239780	0.1157
totChol	0.002960	0.0164
sySBP	0.012674	7.93e-05 ☆☆☆
glucose	0.007855	1.58e-05 ☆☆☆

3.4.2. Random Forest

A random forest model was built using the randomForest tool in the R 4.4.1 program. The prevalence of coronary heart disease served as the model's response variable. "1 234" was chosen as the random seed during the model building phase, which involved setting 500 decision trees to guarantee the reproducibility of the outcomes. The mean decrease in Gini index output by the model was used to rank the importance of eight key variables, thereby identifying the factors that contribute most to the prediction of CHD. The importance ranking results (the Gini index reduction average in brackets) are as follows from high to low: sys BP (126.865258), totChol (116.102463), glucose (110.950006), age (105.808923), cigsPerDay (50.608664), male (18.147900), prevalently (15.406326) and prevalent Stroke (3.700328), as shown in Figure 4.

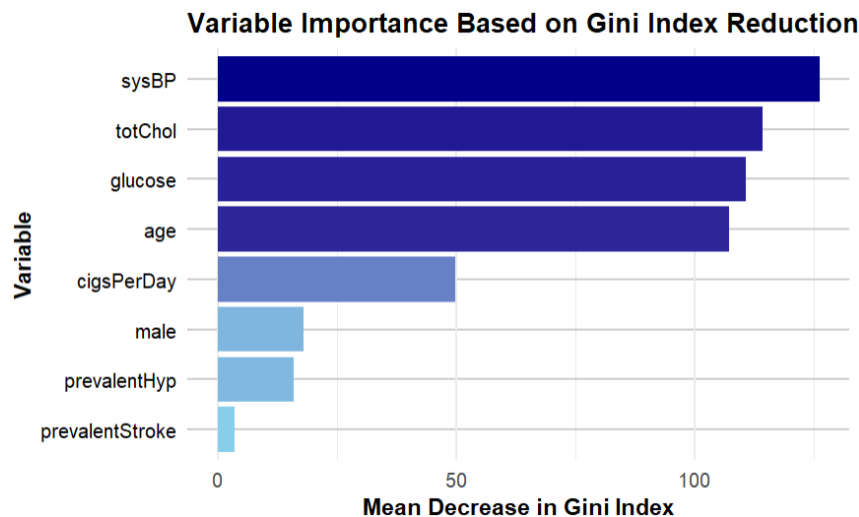


Figure 4. Importance ranking results (Picture credit: Original).

After evaluating the importance of variables, the stepwise random forest method was used to further optimize the model to reduce model complexity and avoid overfitting. Starting from the variable with the highest importance score, the number of variables is gradually reduced to determine the best variable subset. By comparing the OOB estimation error rate under different variable combinations, it is found that when the model contains only the 7 most important variables, the out-of-bag estimation error rate of the model reaches the lowest (as shown in Figure 5), indicating that these 7 variables constitute the best prediction subset. Ranked by importance score, these seven variables are sysBP, totChol, glucose, age, cigsPerDay, male, prevalently, and prevalent Stroke [14].

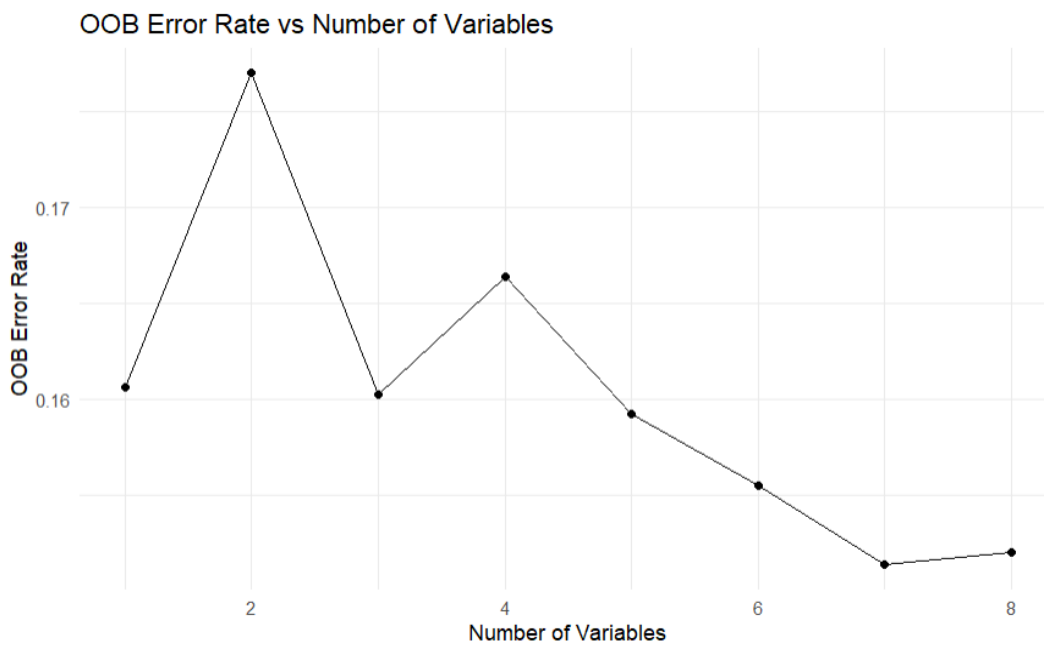


Figure 5. Out-of-bag error rates for models with different numbers of variables (Picture credit: Original).

3.5. Comparative Analysis of Prediction Performance of Prediction Models

3.5.1. Model Performance Comparison

The ROC curve is created to compare the models' classification performance, and the AUC value under the ROC curve for the Random Forest and Logistic Regression models, as well as their accuracy, precision, and recall, are computed for comparison. The major performance indicators for the various models are compiled in Table 8.

Table 8. Key performance indicators of different models

Metric	Logistic Regression	Random Forest
Accuracy	0.856	0.852
Precision	0.858	0.856
Recall	0.995	0.992
F1 Score	0.921	0.919
AUC	0.717	0.615

As shown in Table 8, the accuracy of Logistic Regression and Random Forest is similar, 0.856 and 0.852 respectively. At the same time, the precision and F1 scores of the two models are the same. Nonetheless, the Random Forest model (0.992) does not do as well in recall as the Logistic Regression model (0.995), suggesting that the Logistic Regression model is more flexible in detecting positive samples. Furthermore, the AUC value of Random Forest is 0.615 whereas that of Logistic Regression is 0.717, indicating that Random Forest has a lower overall classification effect and is less effective in differentiating between positive and negative data.

3.5.2. ROC Analyse

This study showed the two models' ROC curves, as seen in Figure 6, to further compare the models' classification performance.

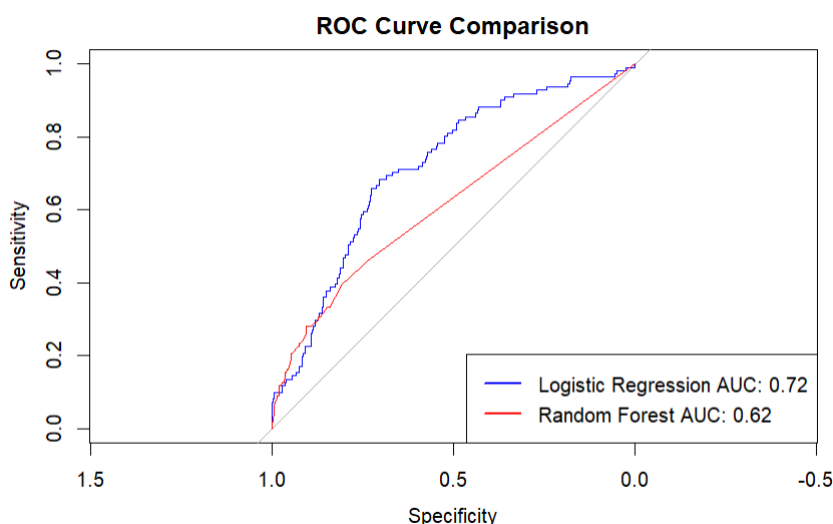


Figure 6. ROC Curve of 2 models (Picture credit: Original).

Figure 6 shows that the ROC curve of Logistic Regression (blue) is above the ROC curve of Random Forest (red), and the AUC value (0.72) is greater than the AUC value of Random Forest (0.62). This shows that Logistic Regression can provide better classification results than Random Forest at different classification thresholds.

3.5.3. Discussion of Model Selection

Comparing various indicators, although Logistic Regression and Random Forest have similar performances in terms of precision, recall, and F1 score, Logistic Regression has a more obvious advantage in terms of AUC. Therefore, the overall classification performance of the Logistic Regression model on this dataset is better than that of the Random Forest, and it is more suitable as the data classification model for this study.

4. Conclusion

This study analyzed cardiovascular research data from residents of Framingham, Massachusetts, and explored the causal relationship between CHD and its related risk factors. The results can not only provide clinicians with more accurate diagnostic tools but also provide a scientific basis for the

formulation of public health policies. This study used machine learning algorithms such as Lasso and Logistic analysis and random forest to screen out characteristic variables that affect CHD, and predicted the incidence of CHD by constructing a risk prediction model. Internal evaluation and internal validation showed that the model had high accuracy and discrimination, showing important clinical practical value.

Numerous characteristics, including age, gender, smoking, hypertension, total cholesterol, systolic and diastolic blood pressure, and total cholesterol level, have been found to be strongly connected with the incidence of congestive heart failure (CHD). Among them, both the Logistic regression model and the random forest model showed that cholesterol and systolic blood pressure had a greater impact, indicating that they have a particularly significant impact on CHD. This is consistent with previous studies and clinical experience, emphasizing the importance of early screening and intervention for these risk factors.

So far, randomized trial evidence to support a multifactorial approach to preventing CHD is relatively limited. However, some data (such as those from the Oslo Diet Heart Study) suggest that this approach is appropriate. In terms of model selection, this study used a multivariate logistic regression model and a random forest model for comparison and finally adopted a multivariate logistic regression model. This model can provide clinicians with a more accurate CHD prediction tool. For example, by using a logistic regression model, doctors can more accurately assess a patient's risk of CHD and thus conduct more effective prevention and intervention measures. In addition, by identifying high-risk groups, lifestyle interventions such as smoking cessation and control of blood pressure and blood sugar can be implemented earlier to reduce CHD morbidity and mortality.

However, the model also has certain limitations. For example, the research data mainly comes from a specific community, Framingham, Massachusetts, and its results may not be fully generalized to other populations. Besides, despite the use of multiple statistical methods for analysis, further research is needed to validate these findings. The results of this study could be improved in the future by incorporating populations from a wider range of ethnic, cultural, and regional backgrounds and by increasing the sample size.

Considering the drawbacks, future studies can investigate more innovative machine learning models, like deep learning and ensemble learning techniques, to enhance the CHD prediction's capacity for generalization and usefulness in real-world applications. In addition, advanced technologies such as genomics and proteomics can be combined to explore the genetic and molecular mechanisms of CHD and provide a more scientific basis for personalized treatment and prevention.

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